

Kasansa Kuya

MECHANICAL ENGINEER | PRODUCT DESIGN | MINING & INDUSTRIAL SYSTEMS

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PROFESSIONAL SUMMARY

Mechanical Engineer with over four years of engineering experience across industrial product development, mining technology, sensor systems, industrial IoT, condition monitoring, prototyping, testing, CAD design, technical documentation, and structural engineering support. Experienced in taking products from concept through design, prototyping, testing, supplier coordination, manufacturing documentation, and deployment support.

Contributed to the development of multiple industrial products, monitoring systems, sensing solutions, and mining-related technologies while working with multidisciplinary teams including mechanical, electronic, firmware, manufacturing, procurement, site, and customer-facing stakeholders. Strong background in engineering problem solving, product development, technical communication, documentation control, and technology-enabled engineering workflows.

Professional focus: product design and development, industrial monitoring systems, mining and harsh-environment equipment, engineering documentation, prototyping, testing, technical problem solving, and technology-enabled engineering systems.

CORE COMPETENCIES

Mechanical & Product Design: Concept development, 3D modelling, design iteration, manufacturability, material selection, enclosure design, assembly planning, prototype development.

Testing & Documentation: Test rigs, validation support, BOMs, drawing registers, manufacturing notes, assembly documentation, technical records, issue tracking.

Software & Data: Python, JavaScript, HTML/CSS, React, Flask basics, Excel, data analysis, AI-assisted workflows, machine learning fundamentals.

Mining & Industrial Systems: Mining-grade equipment, teleremote systems, condition monitoring, sensor integration, fuel monitoring, harsh-environment design, equipment interfaces.

Project Coordination: Supplier communication, procurement support, stakeholder updates, management presentations, multidisciplinary teamwork, deadline management.

CAD & Engineering Tools: SolidWorks, AutoCAD, Revit, Microsoft Word, Excel, PowerPoint, basic Git repository exposure.

PROFESSIONAL EXPERIENCE

Junior Mechanical Engineer

June 2025 - Present

NEROSPEC - Johannesburg, South Africa

- Design and develop multiple industrial products from concept through prototype stage for mining, monitoring, sensing, training, and industrial applications.
- Convert management requirements, product ideas, and technical concepts into manufacturable engineering solutions supported by CAD models, drawings, bills of materials, testing plans, and prototype-development activities.
- Create CAD models, technical drawings, bills of materials, manufacturing notes, supplier information, and engineering documentation for active product-development work.
- Present concepts, design updates, technical constraints, and recommendations to management for review, approval, and further iteration.
- Support prototype development by defining technical requirements, identifying suitable components, coordinating supplier inputs, and resolving manufacturability constraints.
- Develop test setups and validation approaches to support product readiness, reliability, installation planning, and practical deployment.
- Manage several concurrent engineering workstreams while balancing performance, cost, durability, manufacturing feasibility, and delivery timelines.
- Use AI tools and software-assisted workflows to accelerate research, documentation, ideation, troubleshooting, and engineering communication.

Acoustic Machine Health Monitoring Sensor

- Contributed to the development of an acoustic condition-monitoring system designed to assess machine health through vibration and acoustic signatures.
- Supported sensor integration, enclosure design, prototype development, testing, and technical documentation.
- Assisted with machine-condition monitoring concepts aimed at predictive maintenance and equipment reliability improvement.

Hydrogen Pressure Monitoring System

- Participated in development of a pressure-transducer-based monitoring solution for hydrogen-system applications.
- Supported component selection, integration, testing, and documentation activities.
- Considered safety, reliability, and manufacturability requirements during development.

Capacitive Water-Level Sensor

- Contributed to the development of a capacitive sensing solution for liquid-level monitoring applications.
- Assisted with prototyping, validation testing, and mechanical integration activities.
- Supported design refinement and product-development documentation.

Portable Personnel Training System

- Designed and developed a portable demonstration and training platform showcasing machine intervention and monitoring technologies.
- Produced mechanical layouts, component integration solutions, enclosure concepts, and supporting documentation.
- Supported customer demonstrations, technical training, and internal knowledge-transfer activities.

IoT Public Facility Monitoring Platform

- Contributed to development of an IoT-based monitoring system intended for public restroom and facility-management applications.
- Integrated occupancy, door-activation, and consumable-usage monitoring capabilities including soap-dispensing activity.
- Supported hardware integration, testing, documentation, and prototype development.

Junior Mechanical Engineer

Aug 2021 - Sept 2023

NEROSPEC - Johannesburg, South Africa

- Collaborated with multidisciplinary engineering and technical teams on mining technology, IoT-related systems, underground equipment, product design, and site-support projects.
- Provided mechanical engineering support for product design initiatives, prototype builds, jig designs, mounting brackets, equipment enclosures, and manufacturing improvements.
- Conducted engineering investigations to resolve site issues and improve equipment reliability, practicality, manufacturability, and serviceability.
- Maintained engineering documentation including bills of materials, drawing registers, 3D CAD files, manufacturer invoices, project records, and supporting technical files.
- Engaged with mechanical engineering management, assigned project teams, technicians, manufacturers, customer representatives, and site stakeholders.

Fuel Flow Measurement System

- Designed a fuel-flow monitoring device for harsh mining environments with focus on durability, ingress protection, and reliable signal output.
- Developed mechanical packaging using high-strength materials and IP67-oriented design considerations to protect against dust and water ingress.
- Integrated Hall-effect magnetic sensing to detect rotational movement and produce pulse signals for the main control system.
- Supported prototype development, supplier input, documentation, and design refinement for practical site application.

Underground Teleremote Controller

- Worked on mechanical design and integration for an underground teleremote control system used to operate mining machinery from a surface station.
- Supported enclosure design, component mounting, control interface packaging, and integration of mechanical systems with electronics and computers.
- Collaborated with mechanical, radio, electronic, firmware, technician, and mining-focused teams to support a robust electromechanical system.

Underground Meeting / Conferencing System

- Supported the design and implementation of an underground meeting system allowing surface-level mining management to communicate with underground teams.
- Worked with display, camera, computer, speaker, enclosure, and mounting constraints for underground deployment.
- Contributed to mechanical packaging and documentation while collaborating with radio, electronics, firmware, and mining personnel.

Explosion-Proof Battery Box

- Assisted with the design and production readiness of an explosion-proof battery enclosure for mining vehicle use.
- Supported certification-related preparation and redesigned components to improve manufacturability and reduce unnecessary production cost.
- Worked with documentation, component revision, supplier coordination, and practical design constraints for industrial use.

Structural Design Consultant

Oct 2023 - May 2025

Kuyanga Investments Ltd - Ndola, Zambia

- Supported structural design and quality assurance for a commercial shopping mall development.
- Designed reinforced concrete elements including columns, arches, beams, and supporting structural components.
- Translated architectural drawings into structural plans and coordinated with construction teams during project implementation.
- Conducted periodic quality checks to support compliance with design specifications, local building codes, and Zambian building regulations.
- Supported cost-conscious material selection through coordination with teams on locally sourced materials and practical construction constraints.

SELECTED INDEPENDENT TECHNICAL PROJECTS

Portfolio Website and Web Development Projects

- Built a personal engineering and technology portfolio using modern web-development tools, presenting project experience and technical background.
- Developed practical front-end skills in HTML, CSS, JavaScript, React, responsive layouts, user interface components, and deployment workflows.

AI and Software Learning Projects

- Built small software and AI-assisted projects while developing skills in Python, JavaScript, web design, data analysis, and machine learning concepts.
- Used AI tools to support coding, documentation, ideation, structured analysis, and research workflows.

3D Printing and Product Prototyping

- Applied CAD and prototyping knowledge to 3D printing concepts, small product ideas, printable parts, and iterative design workflows.
- Built practical understanding of design constraints, printability, material choice, tolerances, and low-volume manufacturing thinking.

EDUCATION

Bachelor of Engineering - Civil Engineering

2017 - 2021

Jilin Jianzhu University - Changchun, China

- Completed engineering degree with foundation in mathematics, physics, mechanics, applied statistics, structural analysis, fluid dynamics, materials, surveying, and engineering problem solving.
- Studied in an international environment in China, developing cross-cultural communication skills and exposure to Mandarin Chinese.

CERTIFICATIONS AND TECHNICAL LEARNING

- Machine Learning with Python - IBM
- Neural Networks and Deep Learning - DeepLearning.AI
- Introduction to Software Engineering - IBM
- Responsive Web Design - FreeCodeCamp
- AI and Data Analysis fundamentals
- Python for data handling and automation
- JavaScript and front-end web development
- Ongoing self-directed engineering and software learning

TECHNICAL SKILLS

SolidWorks

AutoCAD

Revit

CAD Modelling

Product Design

Prototyping

Design for Manufacture

BOMs

Technical Drawings

Manufacturing Documentation

Testing & Validation

Mining Systems

Condition Monitoring

IoT Exposure

Sensor Integration

Pressure Monitoring

Python

JavaScript

HTML/CSS

React

Flask Basics

Excel

Word

PowerPoint

AI Tools

Data Analysis

LANGUAGES

- English - Fluent

- Afrikaans - Conversational / school-level background
- Mandarin Chinese - Basic to intermediate academic exposure from study in China